

Light Sources and Rays Practice

If an object does not emit light, we can only see it because light bounces off it.

- 1 Light sources produce their own light. If an object does not emit light, we can only see it because light bounces off it.
Decide whether each object is a light source or not.

(a) a fire

(d) the Moon

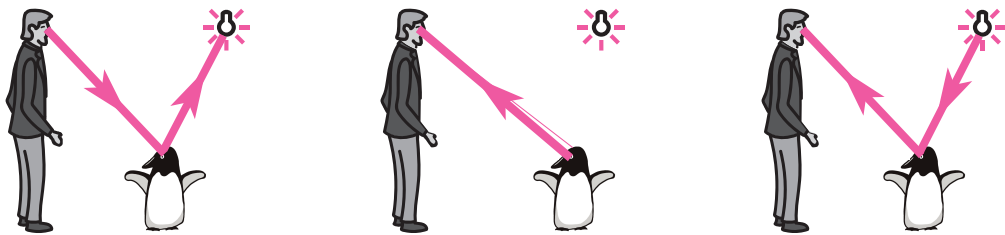
(b) a mirror

(e) a lit match

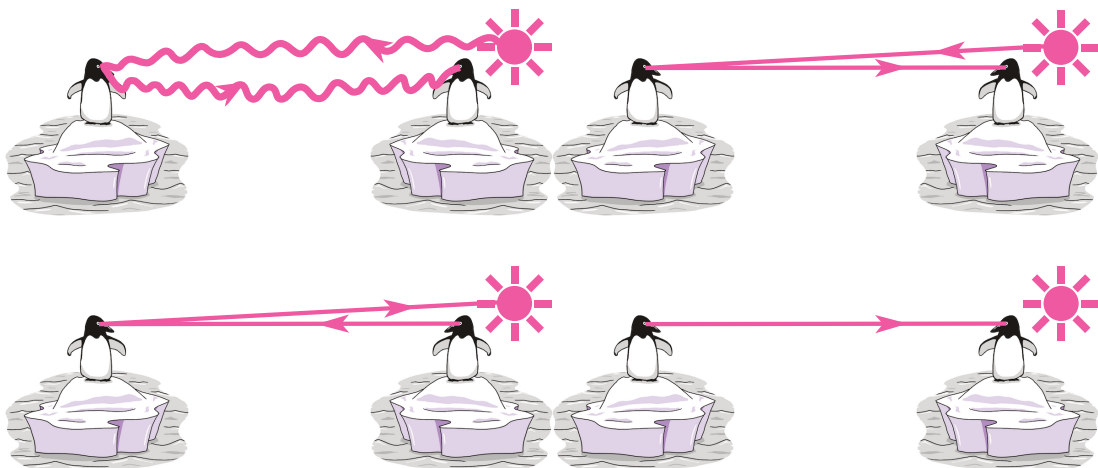
(c) a computer screen

(f) a shiny spoon

- 2 A person can see a penguin in a room with a light bulb. Which of the diagrams below is correct? Circle your answer.



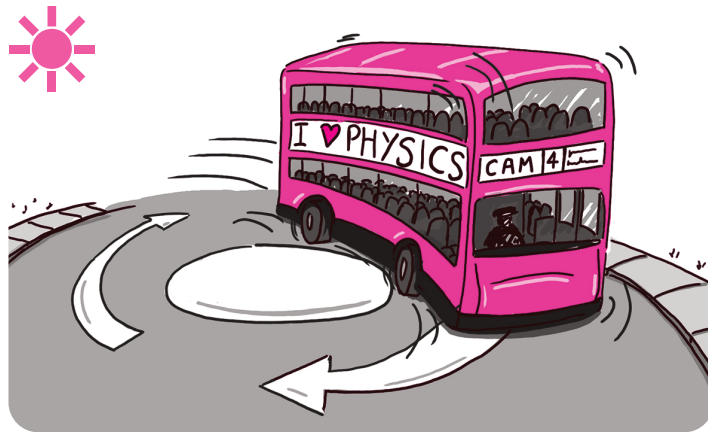
- 3 In the diagrams below, it is sunset and the penguin on the right is looking at its penguin friend on the left. Which of the following diagrams correctly represents how they see the other penguin?



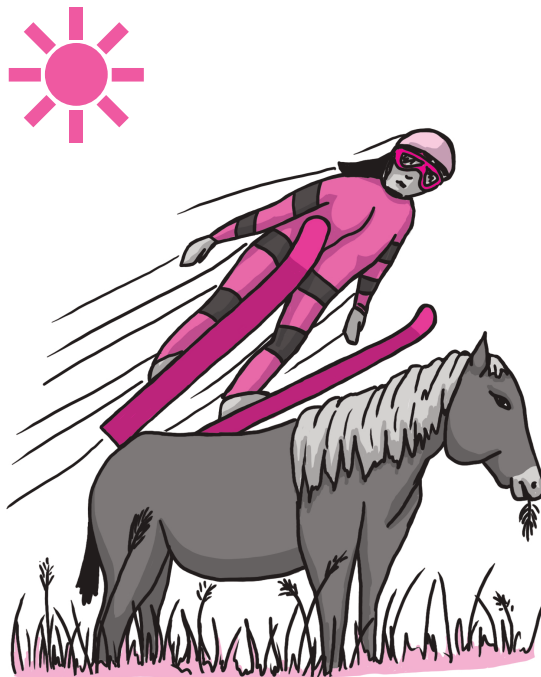
- 4 A student goes to a concert and sees their favourite musician perform! Complete the sentences using the words **reflects**, **student**, **musician**, **spotlight**, **eyes**. You can use a word more than once.

Light from the _____ travels to the _____ and _____ off them. Light then travels from the _____ to the _____ and ends up in their _____.

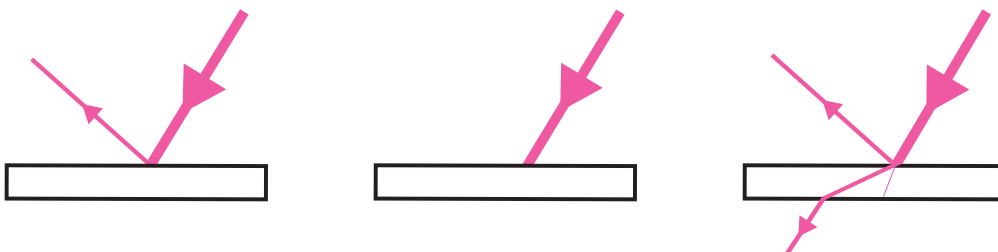
- 5 Draw rays to show how the bus driver sees the roundabout.



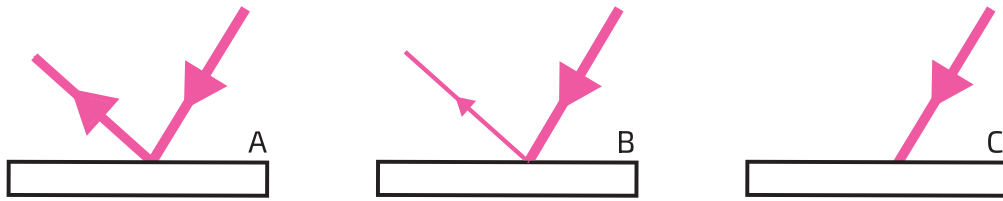
- 6 Draw rays to show how the skier sees the horse's ear.



- 7 Which of the diagrams below correctly shows light reaching an opaque material?



- 8 In these diagrams, the thickness of the line shows the brightness of the light.



(a) Which of these diagrams correctly shows reflection?

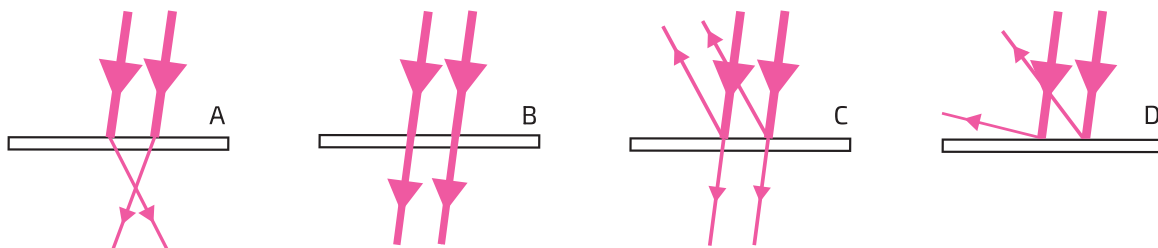
(b) Which of these diagrams correctly shows absorption?

- 9 Complete the sentences using the words **reflected**, **absorbed**, **some**, **no**, **warmer**, **cooler**, **slowly**, **quickly**, **bright**. You can use a word more than once. Some words are not used.

(a) My bike is opaque and painted pale blue. When sunlight shines on it, _____ light goes through it. I can see the bike, because most of the light is _____ by the surface. After the bike has been in sunlight for an hour, it feels much _____. This means that some sunlight was _____.

(b) I paint my bike dark blue. Less light will be _____ off it, and so it won't look as _____. The bike will now warm up more _____ in the sunlight because more light is now being _____.

- 10 Label each diagram as either opaque, translucent or transparent



- 11 Do we need an opaque, transparent or translucent material? Fill in the table.

Description	Opaque, transparent or translucent?
A sleep mask	
A paper globe lampshade for an overly bright lightbulb	
Lenses in reading glasses	